Heterogeneous Federated Learning

Non-IID Data, Client Drift and Heterogeneous Systems

Kingshuk Goswami

Computer Science and Engineering
Dr. B. C. Roy Engineering College, Durgapur

September 20, 2026

From the Previous Work to the New Questions

What was covered previously

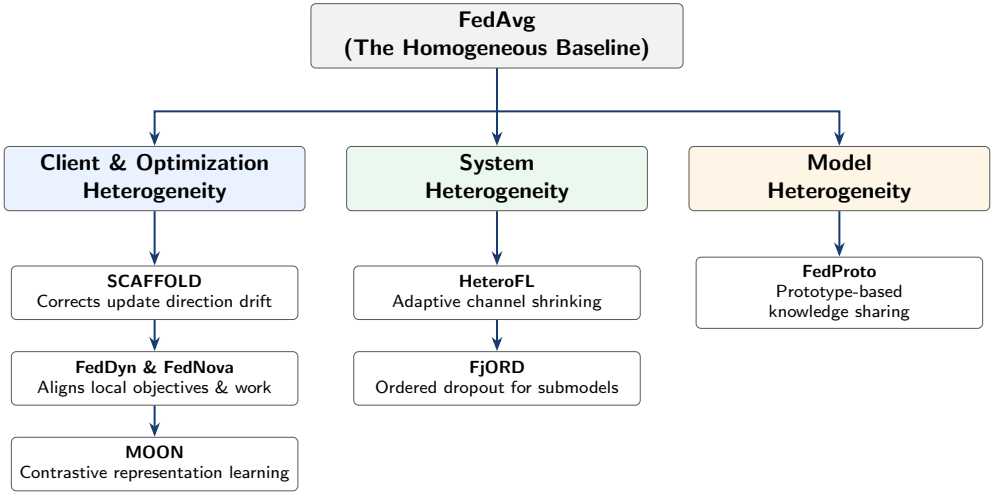
- **FedAvg**: clients train locally and the server averages their models.
- **FedProx**: adds a proximal term to reduce local drift and tolerates partial local work.
- **Personalization**: FedPer, Per-FedAvg and Ditto allow client-specific behavior.
- **Privacy**: raw sensor data remains local, although model updates can still leak information.

Questions that motivate the continuation

1. What if the **direction** of a local gradient is biased?
2. What if clients perform **different amounts of local work**?
3. What if devices have very different **resources**?
4. What if clients cannot use the **same model architecture**?

The continuation moves from "one global model" toward heterogeneous FL.

Navigating the Heterogeneity Landscape



SCAFFOLD

Key Idea: Corrects the client drift caused by non-IID data.

Control Variates

- Maintains global control variate c and local control variate c_i.
- $(c - c_i)$ is the estimate of the total client drift (how different the client's local direction is from the global direction).
- We add $(c - c_i)$ to the client's local gradient g_i to correct the local step before aggregation.
- The local control variate is updated after K local steps using the starting global model x, the new local model y_i, and local learning rate η_l

Gradient Correction

$$\tilde{g}_i = g_i + (c - c_i)$$

Calculating c_i

$$c_i^+ = c_i - c + \frac{x - y_i}{K\eta_l}$$

SCAFFOLD vs. FedProx

SCAFFOLD: changes the direction

FedProx: limits the direction

FedDyn and FedNova

Optimization methods designed to correct objective mismatch and normalize unequal local work.

FedDyn

FedDyn addresses objective inconsistency caused by non-IID client data. It dynamically modifies each client's local objective using a linear correction term and quadratic regularization.
The goal is to make local optimization better aligned with the global objective, reducing client drift.

$$L_k(\theta) - \langle \nabla L_k(\theta_k^{t-1}), \theta \rangle + \frac{\alpha}{2} \|\theta - \theta^{t-1}\|^2$$

- The linear term uses the previous local gradient to modify the local objective, helping compensate for the mismatch between the client's objective and the global objective.
- The quadratic term penalizes moving too far from the previous global model.

FedNova

FedNova addresses objective inconsistency caused by unequal local updates across clients. Clients may perform different numbers of local steps, causing their updates to have different magnitudes.
It normalizes each client's local update before aggregation, separating the direction of progress from the amount of local work.

$$\theta_{t+1} = \theta_t - \eta \, \tau_{\text{eff}} \underbrace{\sum_i p_i \left(\frac{1}{\tau_i} \sum_{k=0}^{\tau_i - 1} g_{i,k} \right)}_{\text{weighted normalized updates}}$$

FedDyn → Corrects the objective function

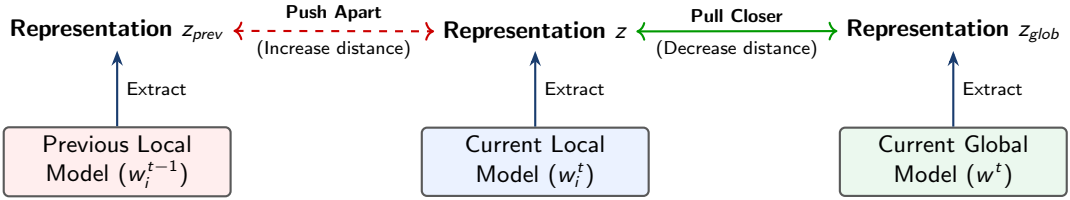
FedNova → Normalizes the update magnitude

MOON

Core Idea

Use **model-level contrastive learning** to correct local training drift caused by non-IID data. It optimizes the current local model's representation to be:

- **Closer** to the representation learned by the **global model** (maximizing agreement).
- **Further** from the representation of the **previous local model** (minimizing agreement).



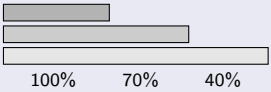
Result: Reduces representation drift and aligns local optimization with the global objective.

HeteroFL and FjORD

System Heterogeneity: Clients have different computational, memory, and bandwidth capabilities, making a single model size impractical for all devices.

HeteroFL

- Generates subnetworks of varying widths by shrinking the hidden channel size by a fixed ratio.
- Distributes these inclusive, nested subnetworks based on client capabilities.
- Smaller local models aggregate with a larger number of clients, stabilizing global aggregation.



FjORD

- Uses **Ordered Dropout** to achieve an ordered, nested representation of knowledge in the DNN.
- Extracts smaller footprint submodels dynamically without the need for retraining.
- Uses self-distillation to transfer knowledge from larger submodels to smaller ones, improving feature extraction.

Same underlying model structure

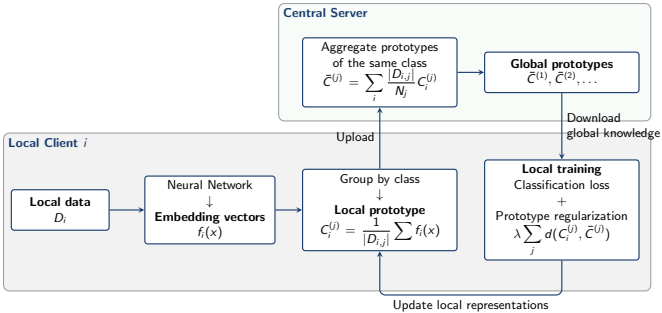
$\Downarrow$

different usable capacities

Both approaches still require all clients to share the same fundamental underlying network architecture.

FedProto

Key Idea: Exchange abstract class prototypes instead of model parameters



Inference Phase

1. New Sample
x

→

2. Feature Extraction
Local model extracts the vector embedding

→

3. Distance Calculation
Compare the embedding with the embeddings of all classes

→

4. Prediction
Class with the smallest distance

Comparing the FL Approaches

Method	Challenge addressed	Main mechanism	What changes?
SCAFFOLD	Client drift	Control variates	Update direction
FedDyn	Objective inconsistency	Dynamic regularization	Local objective
FedNova	Unequal local work	Update normalization	Update magnitude
MOON	Representation drift	Contrastive learning	Representation
HeteroFL	Device capacity	Width-scaled subnetworks	Model size
FjORD	Device capacity	Ordered dropout	Model size
FedProto	Model heterogeneity	Prototype sharing	Knowledge representation

Research Gaps in Federated Learning

01 Multi-dimensional Heterogeneity

Existing methods often target specific forms of heterogeneity: **data distribution, client computation, or model capacity**.
Open question: How can FL jointly handle **data, system and model heterogeneity**?

02 Interpretable Federated Learning

Most FL methods focus primarily on predictive performance, while understanding **what the global model learns and why it makes a prediction** remains challenging.
Open question: How can we achieve **interpretable and reliable FL** across heterogeneous clients?

03 Privacy Beyond Data Locality

Keeping raw data local does not automatically guarantee privacy. **Model updates, gradients, and prototypes** can potentially reveal sensitive information.
Open question: How can FL provide **stronger privacy guarantees**?

References

1. Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian Stich, and Ananda Theertha Suresh, "SCAFFOLD: Stochastic Controlled Averaging for Federated Learning," ICML, 2020.
2. Durmus Alp Emre Acar, Yue Zhao, Ramon Matas Navarro, Matthew Mattina, Paul N. Whatmough, and Venkatesh Saligrama, "Federated Learning Based on Dynamic Regularization," ICLR, 2021.
3. Jianyu Wang, Qinghua Liu, Hao Liang, Gauri Joshi, and H. Vincent Poor, "Tackling the Objective Inconsistency Problem in Heterogeneous Federated Optimization," NeurIPS, 2020.
4. Qinbin Li, Bingsheng He, and Dawn Song, "Model-Contrastive Federated Learning," CVPR, 2021.
5. Enmao Diao, Jie Ding, and Vahid Tarokh, "HeteroFL: Computation and Communication Efficient Federated Learning for Heterogeneous Clients," ICLR, 2021.
6. Samuel Horváth, Stefanos Laskaridis, Mario Almeida, Ilias Leontiadis, Stylianos I. Venieris, and Nicholas D. Lane, "FjORD: Fair and Accurate Federated Learning under Heterogeneous Targets with Ordered Dropout," NeurIPS, 2021.
7. Yue Tan, Guodong Long, Lu Liu, Tianyi Zhou, Qinghua Lu, Jing Jiang, and Chengqi Zhang, "FedProto: Federated Prototype Learning across Heterogeneous Clients," AAAI, 2022.

Thank You